

Estimating Fire Radiative Energy and Rate of Spread in Major Wildfires

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Research Question

How does fire radiative energy released by a fire compare to its rate of spread?

Key Terms

Key Terms

1. **VIIRS**
Visible Infrared Imaging Radiometer Suite – a NOAA–NASA satellite sensor that detects fire “hot spots” and measures instantaneous radiative power (FRP).
2. **MTBS**
Monitoring Trends in Burn Severity – a USFS/USGS program that produces annual burn-severity maps and fire-perimeter shapefiles (incl. dNBR) for large U.S. wildfires.
3. **FRP**
Fire Radiative Power – the instantaneous rate (in MW) at which a fire emits radiant energy, observed by VIIRS.
4. **FRE**
Fire Radiative Energy – the cumulative energy (in MJ / GJ) released by a fire over its duration (the time-integral of FRP).
5. **ROS**
Rate of Spread – the speed of fire expansion, typically in acres per day, derived from perimeter reports (e.g., ICS-209 SITREPs).
6. **dNBR**
Differenced Normalized Burn Ratio – a burn-severity index comparing pre- and post-fire NIR/SWIR reflectance; higher values indicate greater vegetation loss.

Study Fires & Data Sources

Fires Analyzed (n=7)

- Pine Gulch Fire (2020)
- Cameron Peak Fire (2020)
- East Troublesome Fire (2020)
- Watson Creek Fire (2021)
- Caldwell Fire (2021)
- Jennies Peak 1039 Rn (2018)
- Chetco Bar (2017)

Key Data

- **VIIRS FRP** (500 m, twice daily) → Fire Radiative Power
- **ICS-209+** perimeters → daily acreage → ROS (acres / day)
- **MTBS** final perimeters for cross-check

Methods—FRE Calculation

Approach

1. Retrieve VIIRS FRP scans over each incident for its active period
2. Integrate FRP over time (12-hour composites) using:
 - Trapezoidal rule
 - Simpson's rule
 - Midpoint rule
3. Compare integrators → Trapezoidal and Midpoint within <4% difference → choose trapezoid for poster

fire	satellite	FRE_Trapezoid_GJ	FRE_Midpoint_GJ	FRE_Simpson_GJ
Caldwell	ALL	34636255.29	34636255.29	NA
Caldwell	N	4859096.8899	4859096.8899	NA
Caldwell	N20	32324827.5813	32324827.5813	32203386.5677106
East Troublesome	ALL	4263400.7658	4263400.7658	3636691.03153373
East Troublesome	N	4290991.9626	4290991.9626	4059865.93924436
East Troublesome	N20	4203709.8423	4203709.8423	4016676.23334195
Watson Creek	ALL	122649204.7743	122649204.7743	NA
Watson Creek	N	122797821.6102	122797821.6102	NA
Watson Creek	N20	159427439.1891	159427439.1891	NA
Cameron Peak	ALL	9076388.4216	9076388.4216	8506077.52443134
Cameron Peak	N	10150947.4515	10150947.4515	NA
Cameron Peak	N20	12401506.0347	12401506.0347	NA
Jennies Peak 1039 Rn	ALL	6398880.6669	6398880.6669	6326642.22090174
Jennies Peak 1039 Rn	N	3660580.2639	3660580.2639	3590800.46075874
Jennies Peak 1039 Rn	N20	16734.4629	16734.4629	NA
Chetco Bar	ALL	5413321.0863	5413321.0863	5520338.89924632
Chetco Bar	N	5413321.0863	5413321.0863	5520338.89924632
Pine Gulch	ALL	3014146.1082	3014146.1082	3422889.47592202
Pine Gulch	N	3101969.598	3101969.598	3637741.03513876
Pine Gulch	N20	2852454.7995	2852454.7995	3205569.25132769

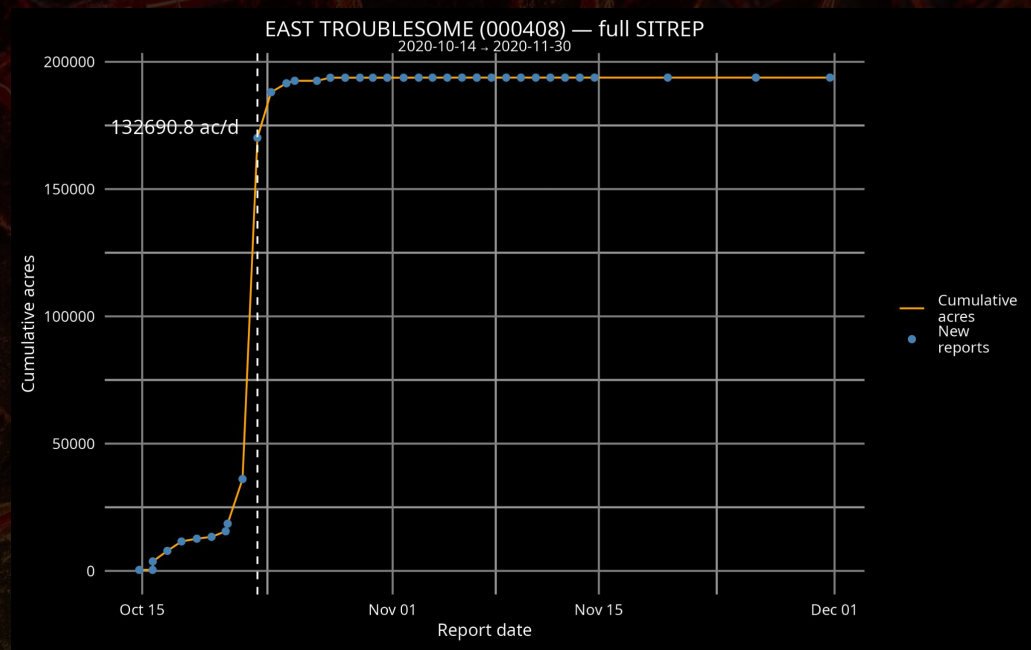
Methods—ROS Calculation

Approach

- Extract daily fire perimeters from ICS-209+
- Compute new area burned between successive perimeters
- $ROS = (\Delta \text{ area}) / (\Delta \text{ days})$

Caveats

- Irregular update intervals → calculate per-day average or per-day at largest peak growth
- Perimeter smoothing by different teams could help



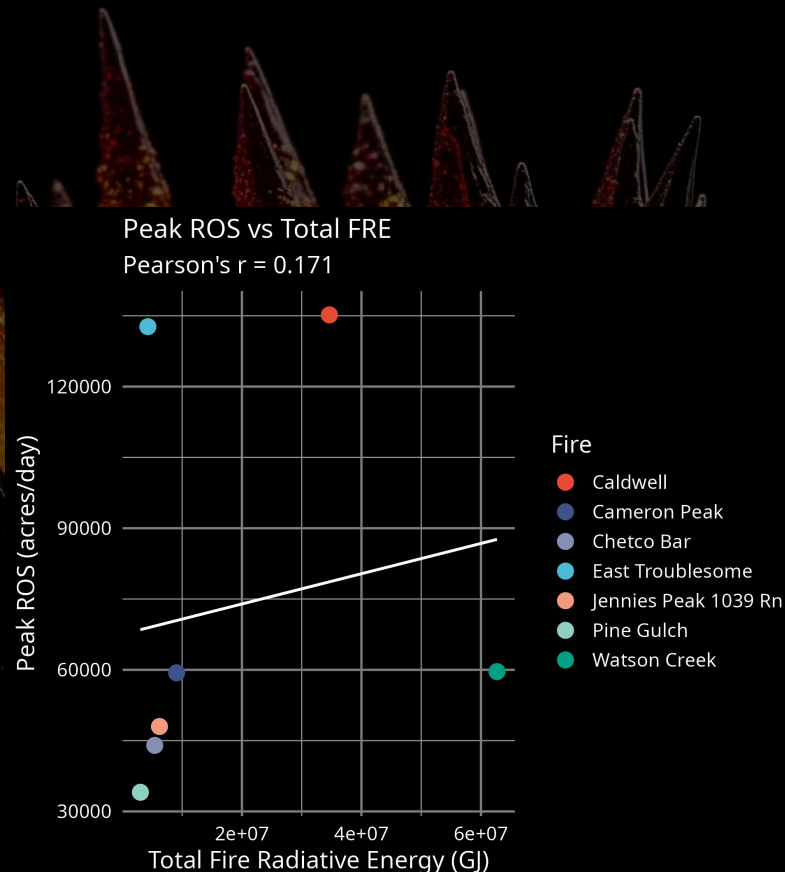
Results—FRE vs ROS

- Strong positive correlation ($R^2 \approx 0.85$) was hypothesized, but data was limited.
- Pearson's r turned out to be 0.171, with linear regression of $R^2 \approx 0.029$.
The data may have been skewed by a couple fires.
- Most fires cluster along 1:1 line

Outliers

- East Troublesome: very high ROS but lower FRE
- Watson Creek: very high FRE but low ROS

More fires would need to be applied to see valid results.



Interpreting the Outliers

Possible explanations for East Troublesome

- VIIRS FRP underestimation in dense smoke or nighttime → missed peaks
- Perimeter reporting lag might inflate ROS
- Less biomass per area → less FRE

Possible explanations for Watson Creek

- VIIRS/ICS-209 timing gaps → FRE captures peaks during times ROS underreports acreage/day
- Small area with heavy fuels: intense burn, limited spread
- Steep, patchy terrain: high energy, slow advance

Field work or historical research of the environment and terrain might be helpful.

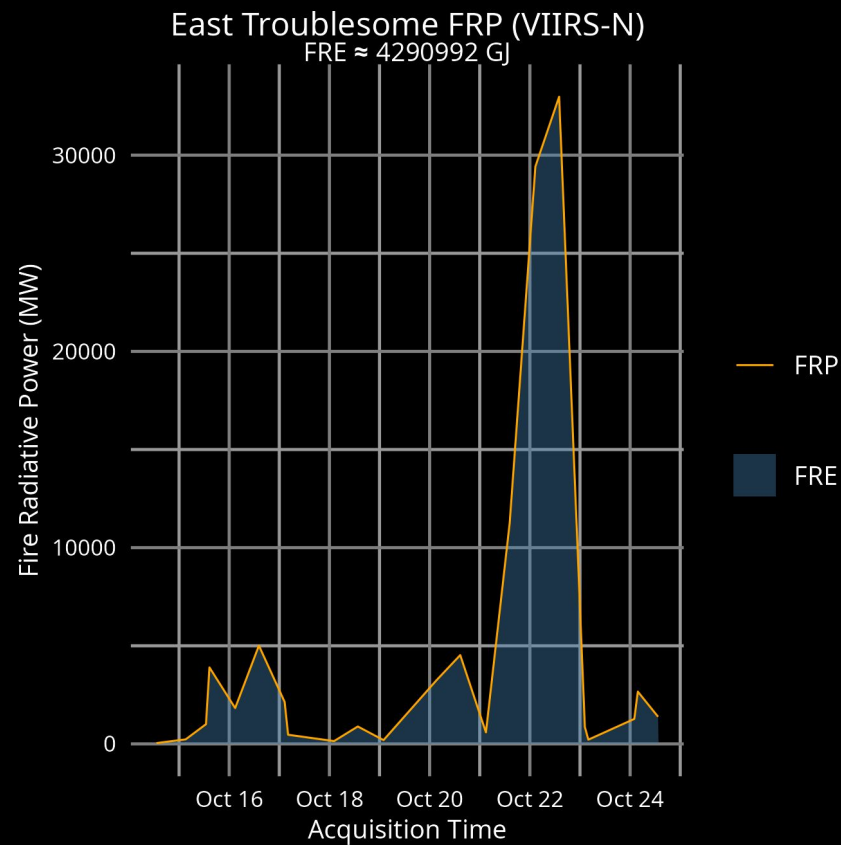
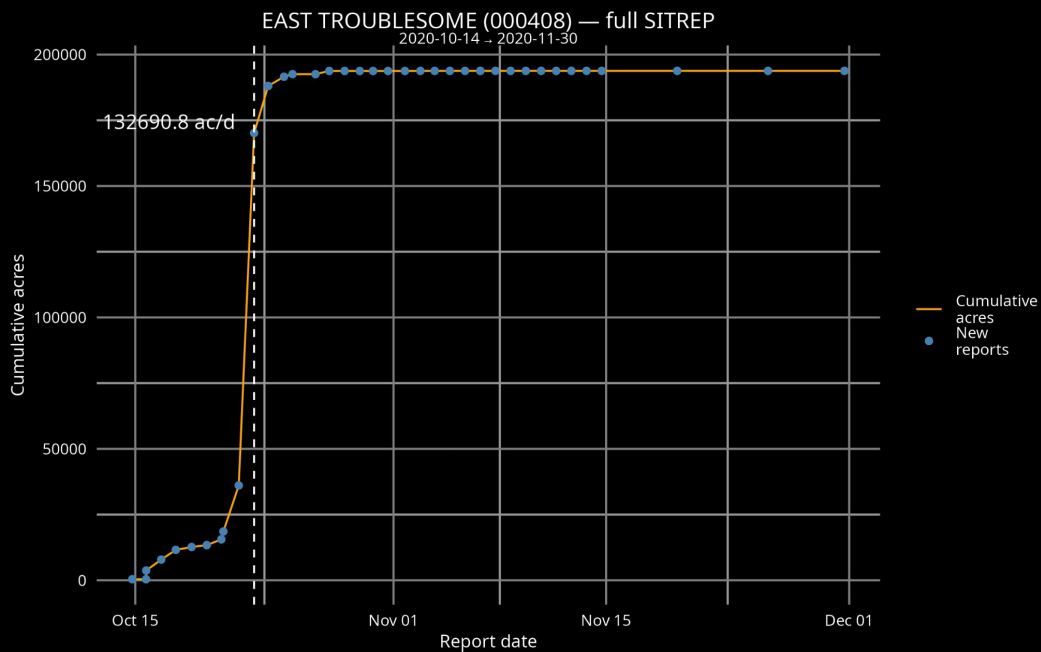
Limitations & Professional Collaboration

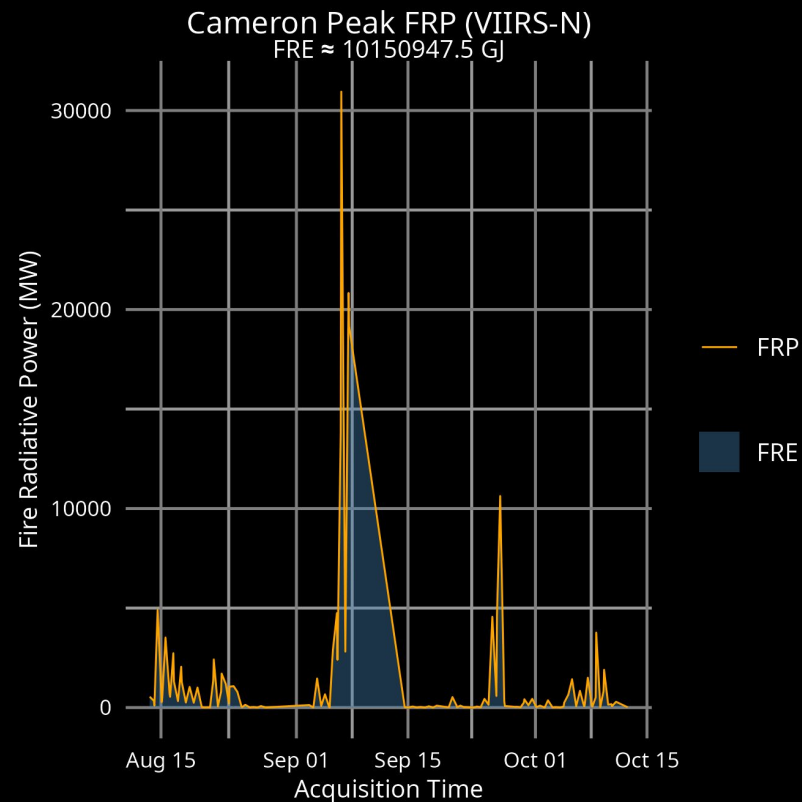
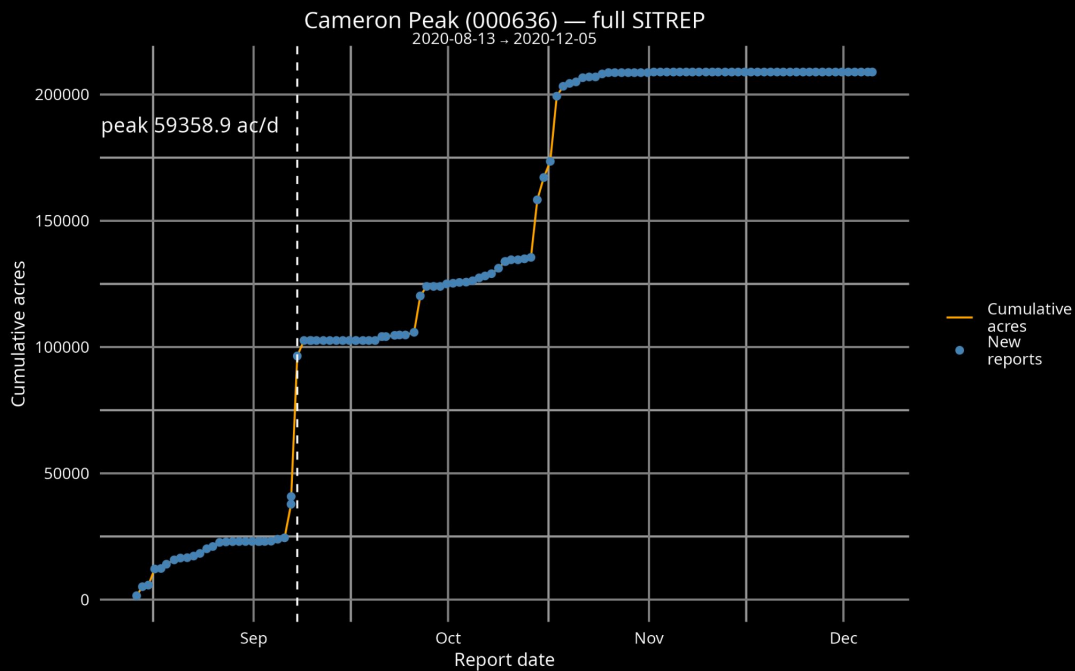
Limitations

- Small sample ($n=7$) → broader fire behavior needed
- VIIRS temporal gaps → missed short-lived peaks

Next Steps

- Expand to 10–20 fires using ICS-209 v3 and making sure VIIRS and MTBS match
- Co-author with remote sensing specialists for publication





Peak fire growth days may be associated with peak releases in fire radiative energy.

Conclusions

- ROS and FRE may be correlated for most fast fires
- The trapezoid rule and midpoint rule give similar results; off by only <4%
- Fuel type & detection gaps might produce outliers (e.g., East Troublesome, Watson Creek)

dNBR vs FRE → Next Steps

What is dNBR?

- dNBR (Differenced Normalized Burn Ratio): change in NIR-SWIR reflectance pre-/post-fire
- A proxy for burn severity at Landsat's 30 m scale

Current status

- Manually downloaded tile for each fire & calculated dNBR
- Found difficulties: cloud masking, exact date matching, calculating proper values

Planned work

1. Refine cloud/cloud-shadow mask (CFMask)
2. Cross-validate dNBR hotspots with ICS/MTBS perimeters and specific pixel regions (MTBS is an interagency mapping program which stands for Monitoring Trends in Burn Severity)
3. Build statistical model: FRE → predicted dNBR

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Check out the QR code for:

- Full code repository & reproducible scripts
- Reference list & further reading
- Earth Lab's wildfire research hub
- My LinkedIn profile for contact and collaboration

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